

Network Working Group
Request for Comments: 4492
Category: Informational

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May 2006

Elliptic Curve Cryptography (ECC) Cipher Suites for Transport Layer Security (TLS)

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Abstract

This document describes new key exchange algorithms based on Elliptic Curve Cryptography (ECC) for the Transport Layer Security (TLS) protocol. In particular, it specifies the use of Elliptic Curve Diffie-Hellman (ECDH) key agreement in a TLS handshake and the use of Elliptic Curve Digital Signature Algorithm (ECDSA) as a new authentication mechanism.

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1. Introduction

Elliptic Curve Cryptography (ECC) is emerging as an attractive public-key cryptosystem, in particular for mobile (i.e., wireless) environments. Compared to currently prevalent cryptosystems such as RSA, ECC offers equivalent security with smaller key sizes. This is illustrated in the following table, based on [18], which gives approximate comparable key sizes for symmetric- and asymmetric-key cryptosystems based on the best-known algorithms for attacking them.

Symmetric		ECC		DH/DSA/RSA
80		163		1024
112		233		2048
128		283		3072
192		409		7680
256		571		15360

Table 1: Comparable Key Sizes (in bits)

Smaller key sizes result in savings for power, memory, bandwidth, and computational cost that make ECC especially attractive for constrained environments.

This document describes additions to TLS to support ECC, applicable both to TLS Version 1.0 [2] and to TLS Version 1.1 [3]. In particular, it defines

- o the use of the Elliptic Curve Diffie-Hellman (ECDH) key agreement scheme with long-term or ephemeral keys to establish the TLS premaster secret, and
- o the use of fixed-ECDH certificates and ECDSA for authentication of TLS peers.

The remainder of this document is organized as follows. Section 2 provides an overview of ECC-based key exchange algorithms for TLS. Section 3 describes the use of ECC certificates for client authentication. TLS extensions that allow a client to negotiate the use of specific curves and point formats are presented in Section 4. Section 5 specifies various data structures needed for an ECC-based handshake, their encoding in TLS messages, and the processing of those messages. Section 6 defines new ECC-based cipher suites and identifies a small subset of these as recommended for all implementations of this specification. Section 7 discusses security considerations. Section 8 describes IANA considerations for the name spaces created by this document. Section 9 gives acknowledgements.

This is followed by the lists of normative and informative references cited in this document, the authors' contact information, and statements on intellectual property rights and copyrights.

Implementation of this specification requires familiarity with TLS [2][3], TLS extensions [4], and ECC [5][6][7][11][17].

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 [1].

2. Key Exchange Algorithms

This document introduces five new ECC-based key exchange algorithms for TLS. All of them use ECDH to compute the TLS premaster secret, and they differ only in the lifetime of ECDH keys (long-term or ephemeral) and the mechanism (if any) used to authenticate them. The derivation of the TLS master secret from the premaster secret and the subsequent generation of bulk encryption/MAC keys and initialization vectors is independent of the key exchange algorithm and not impacted by the introduction of ECC.

The table below summarizes the new key exchange algorithms, which mimic DH_DSS, DHE_DSS, DH_RSA, DHE_RSA, and DH_anon (see [2] and [3]), respectively.

Key Exchange Algorithm -----	Description -----
ECDH_ECDSA	Fixed ECDH with ECDSA-signed certificates.
ECDHE_ECDSA	Ephemeral ECDH with ECDSA signatures.
ECDH_RSA	Fixed ECDH with RSA-signed certificates.
ECDHE_RSA	Ephemeral ECDH with RSA signatures.
ECDH_anon	Anonymous ECDH, no signatures.

Table 2: ECC Key Exchange Algorithms

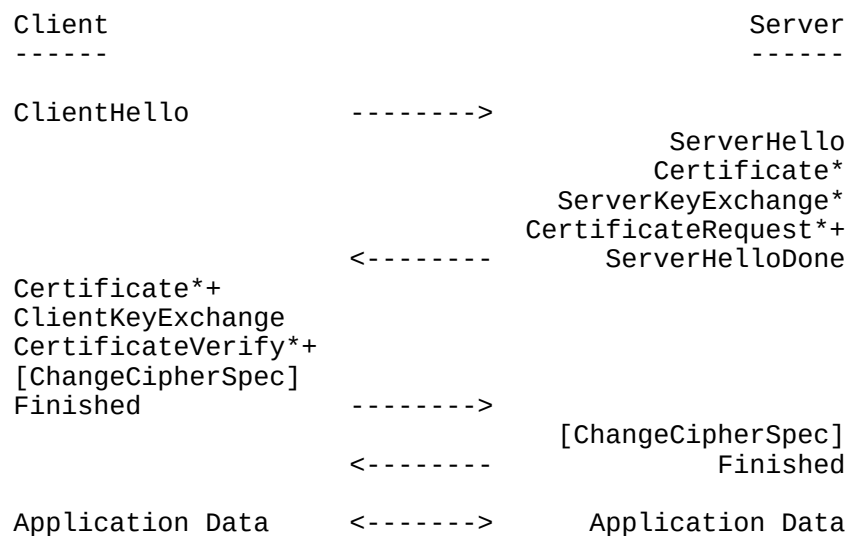
The ECDHE_ECDSA and ECDHE_RSA key exchange mechanisms provide forward secrecy. With ECDHE_RSA, a server can reuse its existing RSA certificate and easily comply with a constrained client's elliptic curve preferences (see Section 4). However, the computational cost

incurred by a server is higher for ECDHE_RSA than for the traditional RSA key exchange, which does not provide forward secrecy.

The ECDH_RSA mechanism requires a server to acquire an ECC certificate, but the certificate issuer can still use an existing RSA key for signing. This eliminates the need to update the keys of trusted certification authorities accepted by TLS clients. The ECDH_ECDSA mechanism requires ECC keys for the server as well as the certification authority and is best suited for constrained devices unable to support RSA.

The anonymous key exchange algorithm does not provide authentication of the server or the client. Like other anonymous TLS key exchanges, it is subject to man-in-the-middle attacks. Implementations of this algorithm SHOULD provide authentication by other means.

Note that there is no structural difference between ECDH and ECDSA keys. A certificate issuer may use X.509 v3 keyUsage and extendedKeyUsage extensions to restrict the use of an ECC public key to certain computations [15]. This document refers to an ECC key as ECDH-capable if its use in ECDH is permitted. ECDSA-capable is defined similarly.



* message is not sent under some conditions
 + message is not sent unless client authentication is desired

Figure 1: Message flow in a full TLS handshake

Figure 1 shows all messages involved in the TLS key establishment protocol (aka full handshake). The addition of ECC has direct impact only on the ClientHello, the ServerHello, the server's Certificate message, the ServerKeyExchange, the ClientKeyExchange, the CertificateRequest, the client's Certificate message, and the CertificateVerify. Next, we describe each ECC key exchange algorithm in greater detail in terms of the content and processing of these messages. For ease of exposition, we defer discussion of client authentication and associated messages (identified with a + in Figure 1) until Section 3 and of the optional ECC-specific extensions (which impact the Hello messages) until Section 4.

2.1. ECDH_ECDSA

In ECDH_ECDSA, the server's certificate MUST contain an ECDH-capable public key and be signed with ECDSA.

A ServerKeyExchange MUST NOT be sent (the server's certificate contains all the necessary keying information required by the client to arrive at the premaster secret).

The client generates an ECDH key pair on the same curve as the server's long-term public key and sends its public key in the ClientKeyExchange message (except when using client authentication algorithm ECDSA_fixed_ECDH or RSA_fixed_ECDH, in which case the modifications from Section 3.2 or Section 3.3 apply).

Both client and server perform an ECDH operation and use the resultant shared secret as the premaster secret. All ECDH calculations are performed as specified in Section 5.10.

2.2. ECDHE_ECDSA

In ECDHE_ECDSA, the server's certificate MUST contain an ECDSA-capable public key and be signed with ECDSA.

The server sends its ephemeral ECDH public key and a specification of the corresponding curve in the ServerKeyExchange message. These parameters MUST be signed with ECDSA using the private key corresponding to the public key in the server's Certificate.

The client generates an ECDH key pair on the same curve as the server's ephemeral ECDH key and sends its public key in the ClientKeyExchange message.

Both client and server perform an ECDH operation (Section 5.10) and use the resultant shared secret as the premaster secret.

2.3. ECDH_RSA

This key exchange algorithm is the same as ECDH_ECDSA except that the server's certificate MUST be signed with RSA rather than ECDSA.

2.4. ECDHE_RSA

This key exchange algorithm is the same as ECDHE_ECDSA except that the server's certificate MUST contain an RSA public key authorized for signing, and that the signature in the ServerKeyExchange message must be computed with the corresponding RSA private key. The server certificate MUST be signed with RSA.

2.5. ECDH_anon

In ECDH_anon, the server's Certificate, the CertificateRequest, the client's Certificate, and the CertificateVerify messages MUST NOT be sent.

The server MUST send an ephemeral ECDH public key and a specification of the corresponding curve in the ServerKeyExchange message. These parameters MUST NOT be signed.

The client generates an ECDH key pair on the same curve as the server's ephemeral ECDH key and sends its public key in the ClientKeyExchange message.

Both client and server perform an ECDH operation and use the resultant shared secret as the premaster secret. All ECDH calculations are performed as specified in Section 5.10.

Note that while the ECDH_ECDSA, ECDHE_ECDSA, ECDH_RSA, and ECDHE_RSA key exchange algorithms require the server's certificate to be signed with a particular signature scheme, this specification (following the similar cases of DH_DSS, DHE_DSS, DH_RSA, and DHE_RSA in [2] and [3]) does not impose restrictions on signature schemes used elsewhere in the certificate chain. (Often such restrictions will be useful, and it is expected that this will be taken into account in certification authorities' signing practices. However, such restrictions are not strictly required in general: Even if it is beyond the capabilities of a client to completely validate a given chain, the client may be able to validate the server's certificate by relying on a trusted certification authority whose certificate appears as one of the intermediate certificates in the chain.)

3. Client Authentication

This document defines three new client authentication mechanisms, each named after the type of client certificate involved: ECDSA_sign, ECDSA_fixed_ECDH, and RSA_fixed_ECDH. The ECDSA_sign mechanism is usable with any of the non-anonymous ECC key exchange algorithms described in Section 2 as well as other non-anonymous (non-ECC) key exchange algorithms defined in TLS [2][3]. The ECDSA_fixed_ECDH and RSA_fixed_ECDH mechanisms are usable with ECDH_ECDSA and ECDH_RSA. Their use with ECDHE_ECDSA and ECDHE_RSA is prohibited because the use of a long-term ECDH client key would jeopardize the forward secrecy property of these algorithms.

The server can request ECC-based client authentication by including one or more of these certificate types in its CertificateRequest message. The server must not include any certificate types that are prohibited for the negotiated key exchange algorithm. The client must check if it possesses a certificate appropriate for any of the methods suggested by the server and is willing to use it for authentication.

If these conditions are not met, the client should send a client Certificate message containing no certificates. In this case, the ClientKeyExchange should be sent as described in Section 2, and the CertificateVerify should not be sent. If the server requires client authentication, it may respond with a fatal handshake failure alert.

If the client has an appropriate certificate and is willing to use it for authentication, it must send that certificate in the client's Certificate message (as per Section 5.6) and prove possession of the private key corresponding to the certified key. The process of determining an appropriate certificate and proving possession is different for each authentication mechanism and described below.

NOTE: It is permissible for a server to request (and the client to send) a client certificate of a different type than the server certificate.

3.1. ECDSA_sign

To use this authentication mechanism, the client MUST possess a certificate containing an ECDSA-capable public key and signed with ECDSA.

The client proves possession of the private key corresponding to the certified key by including a signature in the CertificateVerify message as described in Section 5.8.

3.2. ECDSA_fixed_ECDH

To use this authentication mechanism, the client MUST possess a certificate containing an ECDH-capable public key, and that certificate MUST be signed with ECDSA. Furthermore, the client's ECDH key MUST be on the same elliptic curve as the server's long-term (certified) ECDH key. This might limit use of this mechanism to closed environments. In situations where the client has an ECC key on a different curve, it would have to authenticate using either ECDSA_sign or a non-ECC mechanism (e.g., RSA). Using fixed ECDH for both servers and clients is computationally more efficient than mechanisms providing forward secrecy.

When using this authentication mechanism, the client MUST send an empty ClientKeyExchange as described in Section 5.7 and MUST NOT send the CertificateVerify message. The ClientKeyExchange is empty since the client's ECDH public key required by the server to compute the premaster secret is available inside the client's certificate. The client's ability to arrive at the same premaster secret as the server (demonstrated by a successful exchange of Finished messages) proves possession of the private key corresponding to the certified public key, and the CertificateVerify message is unnecessary.

3.3. RSA_fixed_ECDH

This authentication mechanism is identical to ECDSA_fixed_ECDH except that the client's certificate MUST be signed with RSA.

Note that while the ECDSA_sign, ECDSA_fixed_ECDH, and RSA_fixed_ECDH client authentication mechanisms require the client's certificate to be signed with a particular signature scheme, this specification does not impose restrictions on signature schemes used elsewhere in the certificate chain. (Often such restrictions will be useful, and it is expected that this will be taken into account in certification authorities' signing practices. However, such restrictions are not strictly required in general: Even if it is beyond the capabilities of a server to completely validate a given chain, the server may be able to validate the clients certificate by relying on a trust anchor that appears as one of the intermediate certificates in the chain.)

4. TLS Extensions for ECC

Two new TLS extensions are defined in this specification: (i) the Supported Elliptic Curves Extension, and (ii) the Supported Point Formats Extension. These allow negotiating the use of specific curves and point formats (e.g., compressed vs. uncompressed, respectively) during a handshake starting a new session. These extensions are especially relevant for constrained clients that may

only support a limited number of curves or point formats. They follow the general approach outlined in [4]; message details are specified in Section 5. The client enumerates the curves it supports and the point formats it can parse by including the appropriate extensions in its ClientHello message. The server similarly enumerates the point formats it can parse by including an extension in its ServerHello message.

A TLS client that proposes ECC cipher suites in its ClientHello message SHOULD include these extensions. Servers implementing ECC cipher suites MUST support these extensions, and when a client uses these extensions, servers MUST NOT negotiate the use of an ECC cipher suite unless they can complete the handshake while respecting the choice of curves and compression techniques specified by the client. This eliminates the possibility that a negotiated ECC handshake will be subsequently aborted due to a client's inability to deal with the server's EC key.

The client MUST NOT include these extensions in the ClientHello message if it does not propose any ECC cipher suites. A client that proposes ECC cipher suites may choose not to include these extensions. In this case, the server is free to choose any one of the elliptic curves or point formats listed in Section 5. That section also describes the structure and processing of these extensions in greater detail.

In the case of session resumption, the server simply ignores the Supported Elliptic Curves Extension and the Supported Point Formats Extension appearing in the current ClientHello message. These extensions only play a role during handshakes negotiating a new session.

5. Data Structures and Computations

This section specifies the data structures and computations used by ECC-based key mechanisms specified in Sections 2, 3, and 4. The presentation language used here is the same as that used in TLS [2][3]. Since this specification extends TLS, these descriptions should be merged with those in the TLS specification and any others that extend TLS. This means that enum types may not specify all possible values, and structures with multiple formats chosen with a select() clause may not indicate all possible cases.

5.1. Client Hello Extensions

This section specifies two TLS extensions that can be included with the ClientHello message as described in [4], the Supported Elliptic Curves Extension and the Supported Point Formats Extension.

When these extensions are sent:

The extensions SHOULD be sent along with any ClientHello message that proposes ECC cipher suites.

Meaning of these extensions:

These extensions allow a client to enumerate the elliptic curves it supports and/or the point formats it can parse.

Structure of these extensions:

The general structure of TLS extensions is described in [4], and this specification adds two new types to ExtensionType.

```
enum { elliptic_curves(10), ec_point_formats(11) } ExtensionType;
```

elliptic_curves (Supported Elliptic Curves Extension): Indicates the set of elliptic curves supported by the client. For this extension, the opaque extension_data field contains EllipticCurveList. See Section 5.1.1 for details.

ec_point_formats (Supported Point Formats Extension): Indicates the set of point formats that the client can parse. For this extension, the opaque extension_data field contains ECPointFormatList. See Section 5.1.2 for details.

Actions of the sender:

A client that proposes ECC cipher suites in its ClientHello message appends these extensions (along with any others), enumerating the curves it supports and the point formats it can parse. Clients SHOULD send both the Supported Elliptic Curves Extension and the Supported Point Formats Extension. If the Supported Point Formats Extension is indeed sent, it MUST contain the value 0 (uncompressed) as one of the items in the list of point formats.

Actions of the receiver:

A server that receives a ClientHello containing one or both of these extensions MUST use the client's enumerated capabilities to guide its selection of an appropriate cipher suite. One of the proposed ECC cipher suites must be negotiated only if the server can successfully complete the handshake while using the curves and point formats supported by the client (cf. Sections 5.3 and 5.4).

NOTE: A server participating in an ECDHE-ECDSA key exchange may use different curves for (i) the ECDSA key in its certificate, and (ii) the ephemeral ECDH key in the ServerKeyExchange message. The server must consider the extensions in both cases.

If a server does not understand the Supported Elliptic Curves Extension, does not understand the Supported Point Formats Extension, or is unable to complete the ECC handshake while restricting itself to the enumerated curves and point formats, it MUST NOT negotiate the use of an ECC cipher suite. Depending on what other cipher suites are proposed by the client and supported by the server, this may result in a fatal handshake failure alert due to the lack of common cipher suites.

5.1.1. Supported Elliptic Curves Extension

```
enum {
    sect163k1 (1), sect163r1 (2), sect163r2 (3),
    sect193r1 (4), sect193r2 (5), sect233k1 (6),
    sect233r1 (7), sect239k1 (8), sect283k1 (9),
    sect283r1 (10), sect409k1 (11), sect409r1 (12),
    sect571k1 (13), sect571r1 (14), secp160k1 (15),
    secp160r1 (16), secp160r2 (17), secp192k1 (18),
    secp192r1 (19), secp224k1 (20), secp224r1 (21),
    secp256k1 (22), secp256r1 (23), secp384r1 (24),
    secp521r1 (25),
    reserved (0xFE00..0xFEFF),
    arbitrary_explicit_prime_curves(0xFF01),
    arbitrary_explicit_char2_curves(0xFF02),
    (0xFFFF)
} NamedCurve;
```

sect163k1, etc: Indicates support of the corresponding named curve or class of explicitly defined curves. The named curves defined here are those specified in SEC 2 [13]. Note that many of these curves are also recommended in ANSI X9.62 [7] and FIPS 186-2 [11]. Values 0xFE00 through 0xFEFF are reserved for private use. Values 0xFF01 and 0xFF02 indicate that the client supports arbitrary prime and characteristic-2 curves, respectively (the curve parameters must be encoded explicitly in ECPParameters).

The NamedCurve name space is maintained by IANA. See Section 8 for information on how new value assignments are added.

```
struct {
    NamedCurve elliptic_curve_list<1..2^16-1>
} EllipticCurveList;
```

Items in `elliptic_curve_list` are ordered according to the client's preferences (favorite choice first).

As an example, a client that only supports `secp192r1` (aka NIST P-192; value 19 = 0x0013) and `secp224r1` (aka NIST P-224; value 21 = 0x0015) and prefers to use `secp192r1` would include a TLS extension consisting of the following octets. Note that the first two octets indicate the extension type (Supported Elliptic Curves Extension):

```
00 0A 00 06 00 04 00 13 00 15
```

A client that supports arbitrary explicit characteristic-2 curves (value 0xFF02) would include an extension consisting of the following octets:

```
00 0A 00 04 00 02 FF 02
```

5.1.2. Supported Point Formats Extension

```
enum { uncompressed (0), ansiX962_compressed_prime (1),
      ansiX962_compressed_char2 (2), reserved (248..255)
} ECPointFormat;

struct {
    ECPointFormat ec_point_format_list<1..2^8-1>
} ECPointFormatList;
```

Three point formats are included in the definition of `ECPointFormat` above. The uncompressed point format is the default format in that implementations of this document MUST support it for all of their supported curves. Compressed point formats reduce bandwidth by including only the x-coordinate and a single bit of the y-coordinate of the point. Implementations of this document MAY support the `ansiX962_compressed_prime` and `ansiX962_compressed_char2` formats, where the former applies only to prime curves and the latter applies only to characteristic-2 curves. (These formats are specified in [7].) Values 248 through 255 are reserved for private use.

The `ECPointFormat` name space is maintained by IANA. See Section 8 for information on how new value assignments are added.

Items in `ec_point_format_list` are ordered according to the client's preferences (favorite choice first).

A client that can parse only the uncompressed point format (value 0) includes an extension consisting of the following octets; note that the first two octets indicate the extension type (Supported Point Formats Extension):

```
00 0B 00 02 01 00
```

A client that in the case of prime fields prefers the compressed format (ansiX962_compressed_prime, value 1) over the uncompressed format (value 0), but in the case of characteristic-2 fields prefers the uncompressed format (value 0) over the compressed format (ansiX962_compressed_char2, value 2), may indicate these preferences by including an extension consisting of the following octets:

```
00 0B 00 04 03 01 00 02
```

5.2. Server Hello Extension

This section specifies a TLS extension that can be included with the ServerHello message as described in [4], the Supported Point Formats Extension.

When this extension is sent:

The Supported Point Formats Extension is included in a ServerHello message in response to a ClientHello message containing the Supported Point Formats Extension when negotiating an ECC cipher suite.

Meaning of this extension:

This extension allows a server to enumerate the point formats it can parse (for the curve that will appear in its ServerKeyExchange message when using the ECDHE_ECDSA, ECDHE_RSA, or ECDH_anon key exchange algorithm, or for the curve that is used in the server's public key that will appear in its Certificate message when using the ECDH_ECDSA or ECDH_RSA key exchange algorithm).

Structure of this extension:

The server's Supported Point Formats Extension has the same structure as the client's Supported Point Formats Extension (see Section 5.1.2). Items in `elliptic_curve_list` here are ordered according to the server's preference (favorite choice first). Note that the server may include items that were not found in the client's list (e.g., the server may prefer to receive points in compressed format even when a client cannot parse this format: the same client may nevertheless be capable of outputting points in compressed format).

Actions of the sender:

A server that selects an ECC cipher suite in response to a ClientHello message including a Supported Point Formats Extension appends this extension (along with others) to its ServerHello message, enumerating the point formats it can parse. The Supported Point Formats Extension, when used, MUST contain the value 0 (uncompressed) as one of the items in the list of point formats.

Actions of the receiver:

A client that receives a ServerHello message containing a Supported Point Formats Extension MUST respect the server's choice of point formats during the handshake (cf. Sections 5.6 and 5.7). If no Supported Point Formats Extension is received with the ServerHello, this is equivalent to an extension allowing only the uncompressed point format.

5.3. Server Certificate

When this message is sent:

This message is sent in all non-anonymous ECC-based key exchange algorithms.

Meaning of this message:

This message is used to authentically convey the server's static public key to the client. The following table shows the server certificate type appropriate for each key exchange algorithm. ECC public keys MUST be encoded in certificates as described in Section 5.9.

NOTE: The server's Certificate message is capable of carrying a chain of certificates. The restrictions mentioned in Table 3 apply only to the server's certificate (first in the chain).

Key Exchange Algorithm	Server Certificate Type
-----	-----
ECDH_ECDSA	Certificate MUST contain an ECDH-capable public key. It MUST be signed with ECDSA.
ECDHE_ECDSA	Certificate MUST contain an ECDSA-capable public key. It MUST be signed with ECDSA.
ECDH_RSA	Certificate MUST contain an ECDH-capable public key. It MUST be signed with RSA.
ECDHE_RSA	Certificate MUST contain an RSA public key authorized for use in digital signatures. It MUST be signed with RSA.

Table 3: Server Certificate Types

Structure of this message:

Identical to the TLS Certificate format.

Actions of the sender:

The server constructs an appropriate certificate chain and conveys it to the client in the Certificate message. If the client has used a Supported Elliptic Curves Extension, the public key in the server's certificate MUST respect the client's choice of elliptic curves; in particular, the public key MUST employ a named curve (not the same curve as an explicit curve) unless the client has indicated support for explicit curves of the appropriate type. If the client has used a Supported Point Formats Extension, both the server's public key point and (in the case of an explicit curve) the curve's base point MUST respect the client's choice of point formats. (A server that cannot satisfy these requirements MUST NOT choose an ECC cipher suite in its ServerHello message.)

Actions of the receiver:

The client validates the certificate chain, extracts the server's public key, and checks that the key type is appropriate for the negotiated key exchange algorithm. (A possible reason for a fatal handshake failure is that the client's capabilities for handling elliptic curves and point formats are exceeded; cf. Section 5.1.)

5.4. Server Key Exchange

When this message is sent:

This message is sent when using the ECDHE_ECDSA, ECDHE_RSA, and ECDH_anon key exchange algorithms.

Meaning of this message:

This message is used to convey the server's ephemeral ECDH public key (and the corresponding elliptic curve domain parameters) to the client.

Structure of this message:

```
enum { explicit_prime (1), explicit_char2 (2),  
       named_curve (3), reserved(248..255) } ECCurveType;
```

explicit_prime: Indicates the elliptic curve domain parameters are conveyed verbosely, and the underlying finite field is a prime field.

explicit_char2: Indicates the elliptic curve domain parameters are conveyed verbosely, and the underlying finite field is a characteristic-2 field.

named_curve: Indicates that a named curve is used. This option SHOULD be used when applicable.

Values 248 through 255 are reserved for private use.

The ECCurveType name space is maintained by IANA. See Section 8 for information on how new value assignments are added.

```
struct {  
    opaque a <1..2^8-1>;  
    opaque b <1..2^8-1>;  
} ECCurve;
```

- a, b: These parameters specify the coefficients of the elliptic curve. Each value contains the byte string representation of a field element following the conversion routine in Section 4.3.3 of ANSI X9.62 [7].

```
struct {
    opaque point <1..2^8-1>;
} ECPoint;
```

point: This is the byte string representation of an elliptic curve point following the conversion routine in Section 4.3.6 of ANSI X9.62 [7]. This byte string may represent an elliptic curve point in uncompressed or compressed format; it MUST conform to what the client has requested through a Supported Point Formats Extension if this extension was used.

```
enum { ec_basis_trinomial, ec_basis_pentanomial } ECBasisType;
```

ec_basis_trinomial: Indicates representation of a characteristic-2 field using a trinomial basis.

ec_basis_pentanomial: Indicates representation of a characteristic-2 field using a pentanomial basis.

```
struct {
    ECCurveType curve_type;
    select (curve_type) {
        case explicit_prime:
            opaque prime_p <1..2^8-1>;
            ECCurve curve;
            ECPoint base;
            opaque order <1..2^8-1>;
            opaque cofactor <1..2^8-1>;
        case explicit_char2:
            uint16 m;
            ECBasisType basis;
            select (basis) {
                case ec_trinomial:
                    opaque k <1..2^8-1>;
                case ec_pentanomial:
                    opaque k1 <1..2^8-1>;
                    opaque k2 <1..2^8-1>;
                    opaque k3 <1..2^8-1>;
            };
            ECCurve curve;
            ECPoint base;
            opaque order <1..2^8-1>;
            opaque cofactor <1..2^8-1>;
    };
};
```

```

        case named_curve:
            NamedCurve namedcurve;
        };
    } ECPParameters;

```

curve_type: This identifies the type of the elliptic curve domain parameters.

prime_p: This is the odd prime defining the field F_p .

curve: Specifies the coefficients a and b of the elliptic curve E .

base: Specifies the base point G on the elliptic curve.

order: Specifies the order n of the base point.

cofactor: Specifies the cofactor $h = \#E(F_q)/n$, where $\#E(F_q)$ represents the number of points on the elliptic curve E defined over the field F_q (either F_p or F_{2^m}).

m : This is the degree of the characteristic-2 field F_{2^m} .

k : The exponent k for the trinomial basis representation $x^m + x^k + 1$.

k_1, k_2, k_3 : The exponents for the pentanomial representation $x^m + x^{k_3} + x^{k_2} + x^{k_1} + 1$ (such that $k_3 > k_2 > k_1$).

namedcurve: Specifies a recommended set of elliptic curve domain parameters. All those values of NamedCurve are allowed that refer to a specific curve. Values of NamedCurve that indicate support for a class of explicitly defined curves are not allowed here (they are only permissible in the ClientHello extension); this applies to arbitrary_explicit_prime_curves(0xFF01) and arbitrary_explicit_char2_curves(0xFF02).

```

struct {
    ECPParameters curve_params;
    ECPPoint public;
} ServerECDHParams;

```

curve_params: Specifies the elliptic curve domain parameters associated with the ECDH public key.

public: The ephemeral ECDH public key.

The ServerKeyExchange message is extended as follows.

```
enum { ec_diffie_hellman } KeyExchangeAlgorithm;
```

ec_diffie_hellman: Indicates the ServerKeyExchange message contains an ECDH public key.

```
select (KeyExchangeAlgorithm) {
    case ec_diffie_hellman:
        ServerECDHParams params;
        Signature signed_params;
} ServerKeyExchange;
```

params: Specifies the ECDH public key and associated domain parameters.

signed_params: A hash of the params, with the signature appropriate to that hash applied. The private key corresponding to the certified public key in the server's Certificate message is used for signing.

```
enum { ecdsa } SignatureAlgorithm;

select (SignatureAlgorithm) {
    case ecdsa:
        digitally-signed struct {
            opaque sha_hash[sha_size];
        };
} Signature;
```

```
ServerKeyExchange.signed_params.sha_hash
SHA(ClientHello.random + ServerHello.random +
ServerKeyExchange.params);
```

NOTE: SignatureAlgorithm is "rsa" for the ECDHE_RSA key exchange algorithm and "anonymous" for ECDH_anon. These cases are defined in TLS [2][3]. SignatureAlgorithm is "ecdsa" for ECDHE_ECDSA. ECDSA signatures are generated and verified as described in Section 5.10, and SHA in the above template for sha_hash accordingly may denote a hash algorithm other than SHA-1. As per ANSI X9.62, an ECDSA signature consists of a pair of integers, r and s. The digitally-signed element is encoded as an opaque vector <0..2¹⁶-1>, the contents of which are the DER encoding [9] corresponding to the following ASN.1 notation [8].

```
Ecdsa-Sig-Value ::= SEQUENCE {  
    r      INTEGER,  
    s      INTEGER  
}
```

Actions of the sender:

The server selects elliptic curve domain parameters and an ephemeral ECDH public key corresponding to these parameters according to the ECKAS-DH1 scheme from IEEE 1363 [6]. It conveys this information to the client in the ServerKeyExchange message using the format defined above.

Actions of the receiver:

The client verifies the signature (when present) and retrieves the server's elliptic curve domain parameters and ephemeral ECDH public key from the ServerKeyExchange message. (A possible reason for a fatal handshake failure is that the client's capabilities for handling elliptic curves and point formats are exceeded; cf. Section 5.1.)

5.5. Certificate Request

When this message is sent:

This message is sent when requesting client authentication.

Meaning of this message:

The server uses this message to suggest acceptable client authentication methods.

Structure of this message:

The TLS CertificateRequest message is extended as follows.

```
enum {  
    ecdsa_sign(64), rsa_fixed_ecdh(65),  
    ecdsa_fixed_ecdh(66), (255)  
} ClientCertificateType;
```

ecdsa_sign, etc. Indicates that the server would like to use the corresponding client authentication method specified in Section 3.

Actions of the sender:

The server decides which client authentication methods it would like to use, and conveys this information to the client using the format defined above.

Actions of the receiver:

The client determines whether it has a suitable certificate for use with any of the requested methods and whether to proceed with client authentication.

5.6. Client Certificate

When this message is sent:

This message is sent in response to a CertificateRequest when a client has a suitable certificate and has decided to proceed with client authentication. (Note that if the server has used a Supported Point Formats Extension, a certificate can only be considered suitable for use with the ECDSA_sign, RSA_fixed_ECDH, and ECDSA_fixed_ECDH authentication methods if the public key point specified in it respects the server's choice of point formats. If no Supported Point Formats Extension has been used, a certificate can only be considered suitable for use with these authentication methods if the point is represented in uncompressed point format.)

Meaning of this message:

This message is used to authentically convey the client's static public key to the server. The following table summarizes what client certificate types are appropriate for the ECC-based client authentication mechanisms described in Section 3. ECC public keys must be encoded in certificates as described in Section 5.9.

NOTE: The client's Certificate message is capable of carrying a chain of certificates. The restrictions mentioned in Table 4 apply only to the client's certificate (first in the chain).

Client Authentication Method -----	Client Certificate Type -----
ECDSA_sign	Certificate MUST contain an ECDSA-capable public key and be signed with ECDSA.
ECDSA_fixed_ECDH	Certificate MUST contain an ECDH-capable public key on the same elliptic curve as the server's long-term ECDH key. This certificate MUST be signed with ECDSA.
RSA_fixed_ECDH	Certificate MUST contain an ECDH-capable public key on the same elliptic curve as the server's long-term ECDH key. This certificate MUST be signed with RSA.

Table 4: Client Certificate Types

Structure of this message:

Identical to the TLS client Certificate format.

Actions of the sender:

The client constructs an appropriate certificate chain, and conveys it to the server in the Certificate message.

Actions of the receiver:

The TLS server validates the certificate chain, extracts the client's public key, and checks that the key type is appropriate for the client authentication method.

5.7. Client Key Exchange

When this message is sent:

This message is sent in all key exchange algorithms. If client authentication with ECDSA_fixed_ECDH or RSA_fixed_ECDH is used, this message is empty. Otherwise, it contains the client's ephemeral ECDH public key.

Meaning of the message:

This message is used to convey ephemeral data relating to the key exchange belonging to the client (such as its ephemeral ECDH public key).

Structure of this message:

The TLS ClientKeyExchange message is extended as follows.

```
enum { implicit, explicit } PublicValueEncoding;
```

implicit, explicit: For ECC cipher suites, this indicates whether the client's ECDH public key is in the client's certificate ("implicit") or is provided, as an ephemeral ECDH public key, in the ClientKeyExchange message ("explicit"). (This is "explicit" in ECC cipher suites except when the client uses the ECDSA_fixed_ECDH or RSA_fixed_ECDH client authentication mechanism.)

```
struct {
    select (PublicValueEncoding) {
        case implicit: struct { };
        case explicit: ECPoint ecdh_Yc;
    } ecdh_public;
} ClientECDiffieHellmanPublic;
```

ecdh_Yc: Contains the client's ephemeral ECDH public key as a byte string ECPoint.point, which may represent an elliptic curve point in uncompressed or compressed format. Here, the format MUST conform to what the server has requested through a Supported Point Formats Extension if this extension was used, and MUST be uncompressed if this extension was not used.

```
struct {
    select (KeyExchangeAlgorithm) {
        case ec_diffie_hellman: ClientECDiffieHellmanPublic;
    } exchange_keys;
} ClientKeyExchange;
```

Actions of the sender:

The client selects an ephemeral ECDH public key corresponding to the parameters it received from the server according to the ECKAS-DH1 scheme from IEEE 1363 [6]. It conveys this information to the client in the ClientKeyExchange message using the format defined above.

Actions of the receiver:

The server retrieves the client's ephemeral ECDH public key from the ClientKeyExchange message and checks that it is on the same elliptic curve as the server's ECDH key.

5.8. Certificate Verify

When this message is sent:

This message is sent when the client sends a client certificate containing a public key usable for digital signatures, e.g., when the client is authenticated using the ECDSA_sign mechanism.

Meaning of the message:

This message contains a signature that proves possession of the private key corresponding to the public key in the client's Certificate message.

Structure of this message:

The TLS CertificateVerify message and the underlying Signature type are defined in [2] and [3], and the latter is extended here in Section 5.4. For the ecdsa case, the signature field in the CertificateVerify message contains an ECDSA signature computed over handshake messages exchanged so far, exactly similar to CertificateVerify with other signing algorithms in [2] and [3]:

```
CertificateVerify.signature.sha_hash
    SHA(handshake_messages);
```

ECDSA signatures are computed as described in Section 5.10, and SHA in the above template for sha_hash accordingly may denote a hash algorithm other than SHA-1. As per ANSI X9.62, an ECDSA signature consists of a pair of integers, r and s. The digitally-signed element is encoded as an opaque vector <0..2¹⁶-1>, the contents of which are the DER encoding [9] corresponding to the following ASN.1 notation [8].

```
Ecdsa-Sig-Value ::= SEQUENCE {
    r      INTEGER,
    s      INTEGER
}
```

Actions of the sender:

The client computes its signature over all handshake messages sent or received starting at client hello and up to but not including this message. It uses the private key corresponding to its certified public key to compute the signature, which is conveyed in the format defined above.

Actions of the receiver:

The server extracts the client's signature from the CertificateVerify message, and verifies the signature using the public key it received in the client's Certificate message.

5.9. Elliptic Curve Certificates

X.509 certificates containing ECC public keys or signed using ECDSA MUST comply with [14] or another RFC that replaces or extends it. Clients SHOULD use the elliptic curve domain parameters recommended in ANSI X9.62 [7], FIPS 186-2 [11], and SEC 2 [13].

5.10. ECDH, ECDSA, and RSA Computations

All ECDH calculations (including parameter and key generation as well as the shared secret calculation) are performed according to [6] using the ECKAS-DH1 scheme with the identity map as key derivation function (KDF), so that the premaster secret is the x-coordinate of the ECDH shared secret elliptic curve point represented as an octet string. Note that this octet string (Z in IEEE 1363 terminology) as output by FE2OSP, the Field Element to Octet String Conversion Primitive, has constant length for any given field; leading zeros found in this octet string MUST NOT be truncated.

(Note that this use of the identity KDF is a technicality. The complete picture is that ECDH is employed with a non-trivial KDF because TLS does not directly use the premaster secret for anything other than for computing the master secret. As of TLS 1.0 [2] and 1.1 [3], this means that the MD5- and SHA-1-based TLS PRF serves as a KDF; it is conceivable that future TLS versions or new TLS extensions introduced in the future may vary this computation.)

All ECDSA computations MUST be performed according to ANSI X9.62 [7] or its successors. Data to be signed/verified is hashed, and the result run directly through the ECDSA algorithm with no additional hashing. The default hash function is SHA-1 [10], and sha_size (see Sections 5.4 and 5.8) is 20. However, an alternative hash function, such as one of the new SHA hash functions specified in FIPS 180-2 [10], may be used instead if the certificate containing the EC public

key explicitly requires use of another hash function. (The mechanism for specifying the required hash function has not been standardized, but this provision anticipates such standardization and obviates the need to update this document in response. Future PKIX RFCs may choose, for example, to specify the hash function to be used with a public key in the parameters field of subjectPublicKeyInfo.)

All RSA signatures must be generated and verified according to PKCS#1 [12] block type 1.

6. Cipher Suites

The table below defines new ECC cipher suites that use the key exchange algorithms specified in Section 2.

CipherSuite	TLS_ECDH_ECDSA_WITH_NULL_SHA	= { 0xC0, 0x01 }
CipherSuite	TLS_ECDH_ECDSA_WITH_RC4_128_SHA	= { 0xC0, 0x02 }
CipherSuite	TLS_ECDH_ECDSA_WITH_3DES_EDE_CBC_SHA	= { 0xC0, 0x03 }
CipherSuite	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA	= { 0xC0, 0x04 }
CipherSuite	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA	= { 0xC0, 0x05 }
CipherSuite	TLS_ECDHE_ECDSA_WITH_NULL_SHA	= { 0xC0, 0x06 }
CipherSuite	TLS_ECDHE_ECDSA_WITH_RC4_128_SHA	= { 0xC0, 0x07 }
CipherSuite	TLS_ECDHE_ECDSA_WITH_3DES_EDE_CBC_SHA	= { 0xC0, 0x08 }
CipherSuite	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA	= { 0xC0, 0x09 }
CipherSuite	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA	= { 0xC0, 0x0A }
CipherSuite	TLS_ECDH_RSA_WITH_NULL_SHA	= { 0xC0, 0x0B }
CipherSuite	TLS_ECDH_RSA_WITH_RC4_128_SHA	= { 0xC0, 0x0C }
CipherSuite	TLS_ECDH_RSA_WITH_3DES_EDE_CBC_SHA	= { 0xC0, 0x0D }
CipherSuite	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA	= { 0xC0, 0x0E }
CipherSuite	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA	= { 0xC0, 0x0F }
CipherSuite	TLS_ECDHE_RSA_WITH_NULL_SHA	= { 0xC0, 0x10 }
CipherSuite	TLS_ECDHE_RSA_WITH_RC4_128_SHA	= { 0xC0, 0x11 }
CipherSuite	TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA	= { 0xC0, 0x12 }
CipherSuite	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA	= { 0xC0, 0x13 }
CipherSuite	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA	= { 0xC0, 0x14 }
CipherSuite	TLS_ECDH_anon_WITH_NULL_SHA	= { 0xC0, 0x15 }
CipherSuite	TLS_ECDH_anon_WITH_RC4_128_SHA	= { 0xC0, 0x16 }
CipherSuite	TLS_ECDH_anon_WITH_3DES_EDE_CBC_SHA	= { 0xC0, 0x17 }
CipherSuite	TLS_ECDH_anon_WITH_AES_128_CBC_SHA	= { 0xC0, 0x18 }
CipherSuite	TLS_ECDH_anon_WITH_AES_256_CBC_SHA	= { 0xC0, 0x19 }

Table 5: TLS ECC cipher suites

The key exchange method, cipher, and hash algorithm for each of these cipher suites are easily determined by examining the name. Ciphers (other than AES ciphers) and hash algorithms are defined in [2] and [3]. AES ciphers are defined in [19].

Server implementations SHOULD support all of the following cipher suites, and client implementations SHOULD support at least one of them: TLS_ECDH_ECDSA_WITH_3DES_EDE_CBC_SHA, TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA, and TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA.

7. Security Considerations

Security issues are discussed throughout this memo.

For TLS handshakes using ECC cipher suites, the security considerations in appendices D.2 and D.3 of [2] and [3] apply accordingly.

Security discussions specific to ECC can be found in [6] and [7]. One important issue that implementers and users must consider is elliptic curve selection. Guidance on selecting an appropriate elliptic curve size is given in Table 1.

Beyond elliptic curve size, the main issue is elliptic curve structure. As a general principle, it is more conservative to use elliptic curves with as little algebraic structure as possible. Thus, random curves are more conservative than special curves such as Koblitz curves, and curves over F_p with p random are more conservative than curves over F_p with p of a special form (and curves over F_p with p random might be considered more conservative than curves over F_{2^m} as there is no choice between multiple fields of similar size for characteristic 2). Note, however, that algebraic structure can also lead to implementation efficiencies, and implementers and users may, therefore, need to balance conservatism against a need for efficiency. Concrete attacks are known against only very few special classes of curves, such as supersingular curves, and these classes are excluded from the ECC standards that this document references [6], [7].

Another issue is the potential for catastrophic failures when a single elliptic curve is widely used. In this case, an attack on the elliptic curve might result in the compromise of a large number of keys. Again, this concern may need to be balanced against efficiency and interoperability improvements associated with widely-used curves. Substantial additional information on elliptic curve choice can be found in [5], [6], [7], and [11].

Implementers and users must also consider whether they need forward secrecy. Forward secrecy refers to the property that session keys are not compromised if the static, certified keys belonging to the server and client are compromised. The ECDHE_ECDSA and ECDHE_RSA key exchange algorithms provide forward secrecy protection in the event of server key compromise, while ECDH_ECDSA and ECDH_RSA do not. Similarly, if the client is providing a static, certified key, ECDSA_sign client authentication provides forward secrecy protection in the event of client key compromise, while ECDSA_fixed_ECDH and RSA_fixed_ECDH do not. Thus, to obtain complete forward secrecy protection, ECDHE_ECDSA or ECDHE_RSA must be used for key exchange, with ECDSA_sign used for client authentication if necessary. Here again the security benefits of forward secrecy may need to be balanced against the improved efficiency offered by other options.

8. IANA Considerations

This document describes three new name spaces for use with the TLS protocol:

- o NamedCurve (Section 5.1)
- o ECPointFormat (Section 5.1)
- o ECCurveType (Section 5.4)

For each name space, this document defines the initial value assignments and defines a range of 256 values (NamedCurve) or eight values (ECPointFormat and ECCurveType) reserved for Private Use. Any additional assignments require IETF Consensus action [16].

9. Acknowledgements

The authors wish to thank Bill Anderson and Tim Dierks.

10. References

10.1. Normative References

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10.2. Informative References

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Appendix A. Equivalent Curves (Informative)

All of the NIST curves [11] and several of the ANSI curves [7] are equivalent to curves listed in Section 5.1.1. In the following table, multiple names in one row represent aliases for the same curve.

Curve names chosen by different standards organizations		
SECG	ANSI X9.62	NIST
sect163k1		NIST K-163
sect163r1		
sect163r2		NIST B-163
sect193r1		
sect193r2		
sect233k1		NIST K-233
sect233r1		NIST B-233
sect239k1		
sect283k1		NIST K-283
sect283r1		NIST B-283
sect409k1		NIST K-409
sect409r1		NIST B-409
sect571k1		NIST K-571
sect571r1		NIST B-571
secp160k1		
secp160r1		
secp160r2		
secp192k1		
secp192r1	prime192v1	NIST P-192
secp224k1		
secp224r1		NIST P-224
secp256k1		
secp256r1	prime256v1	NIST P-256
secp384r1		NIST P-384
secp521r1		NIST P-521

Table 6: Equivalent curves defined by SECG, ANSI, and NIST

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Acknowledgement

Funding for the RFC Editor function is provided by the IETF Administrative Support Activity (IASA).